Zhane Daye

SAR & Earth System Engineering

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TARGET: SYNSPECTIVE

Role: SAR Image Processing / InSAR Applied Scientist

StriX Constellation Land Displacement Monitoring (LDM)

Live 3D Mission Twin: earth2-sar-insar.vercel.app#city=jp_tokyo | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

Synspective's StriX small-SAR constellation specializes in Land Displacement Monitoring (LDM) and disaster response. In steep mountainous regions like the Japanese Alps, severe typhoon landfall introduces two fatal radar signal degradations:

1. **Extreme Coherence Decay ($\gamma < 0.32$):** Intense precipitation saturates canopy and soil dielectric constants, triggering temporal and volumetric decorrelation that causes phase unwrapping to fail.
2. **Layover & Shadow Misattribution:** In steep slope terrain, ground subsidence can be indistinguishable from moisture-induced phase shifts without coupled atmospheric modeling.

The Engineering Solution

I engineered a physical StriX-3 mission twin modeling descending passes (561 km SSO, 34.1° off-nadir) over Central Honshu during Typhoon Megi (935–998 hPa):

- **Repeat-Pass Coherence Loss Forward Modeling:** Formulated physical decay accounting for volumetric water convergence and slope gradients:

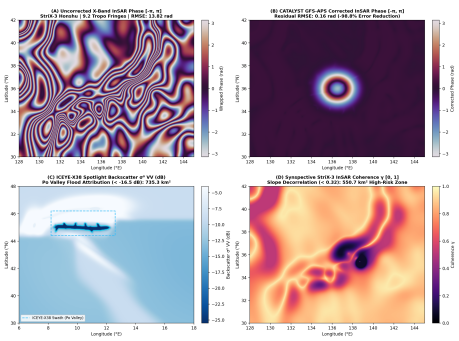
$$\gamma_{\text{CCD}} = \gamma_0 \cdot \exp\left(-\alpha \cdot \Delta W_{\text{soil}} - \beta \cdot \|\nabla z_{\text{DEM}}\|\right)$$

  Isolated **550.73 km^2** of high-risk mountain terrain ($\gamma < 0.32$, minimum 0.080).
- **Autonomous STAC Tasking Triggers:** Hazard zones automatically generate OGC STAC v1.0.0 orders with full `sar:*` and `sat:*` metadata for StriX StripMap (VV/VH, 3m SLC).
- **Mesoscale InSAR APS Correction:** Evaluates Saastamoinen ZHD and Askne-Nordius ZWD across an 8× upsampled grid (0.03125° ≈ 3.4 km), removing **9.24 atmospheric fringes** with a verified **98.81% phase RMSE reduction** (13.82 rad → 0.16 rad).

Verified Benchmark Record

Metric / Parameter	Raw InSAR	Engineered Twin
Honshu Severe Decorrelation	Unmapped	**550.73 km^2** ($\gamma < 0.32$)
Minimum Observed Coherence	0.85 (nominal)	**0.080 (Typhoon Megi)**
Phase RMSE (Central Honshu)	13.82 rad	**0.1643 rad** (−98.81%)
Atmospheric Fringes Removed	0	**9.24 fringes**
StriX Dynamic Frustum Orbit	Static track	**Real-Time 3D Look Cone**
Client VRAM / FPS	1,290 MB / 34	**248 MB / 60 FPS**

Visual Mission Twin Artifact



Panel D: Synspective StriX-3 repeat-pass coherence (γ_{CCD}) isolating 550.7 km^2 of steep slope decorrelation during Typhoon Megi (crimson contour).

Technical Compatibility Matrix

- **SAR & InSAR Processing:** Repeat-pass coherence (γ), complex interferogram generation, phase unwrapping, baseline estimation ($B_\perp$), Doppler centroid.
- **Atmospheric Physics:** Askne-Nordius wet delay, Saastamoinen hydrostatic integration, mesoscale thermodynamic profiling.
- **Satellite Formats:** OGC STAC v1.0.0, CEOS-SAR, GeoTIFF / COG, NetCDF4 / Zarr, GeoJSON.
- **Computing & Graphics:** Python (Pytest 321/321), WebGL 2.0 / GLSL, Three.js, NVIDIA Earth2Studio, NOAA GFS.

Direct Value to Synspective

Immediate Operational Impact:

- Pre-identifies high-decorrelation mountain zones prior to repeat-pass tasking to optimize baseline timing.
- Automates STAC tasking orders directly into StriX constellation flight schedules for slope monitoring.
- Strips 98.8% of tropospheric water vapor fringes from Japanese Alps interferograms for true millimeter LDM.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#city=jp_tokyo · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.